



CAN WE TRUST GRAMMAR-FREE AI? WHAT WE LOSE WHEN WE DROP FORMAL PARSERS

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Why this matters

- ▶ Chatbots sound confident, even when they're wrong.
- ▶ Some applications (law, medicine, grading, code) need models that fail safely and predictably, not randomly
- ▶ Classic parsers can be proven correct; modern language models cannot.

What guarantees of correctness, completeness, and verifiability are lost when automata-based grammar parsers are replaced by probabilistic language models?

What we already knew

1 Structural limits

- ▶ Neural models like LSTMs and Transformers do not have as much formal power as pushdown automata; under realistic assumptions they cannot handle every context-free grammar (Ackerman & Cybenko 2020).
- ▶ In practice, that means they can mimic many patterns, but they cannot give the same rock-solid guarantee that all and only grammatical sentences will be accepted.

2 Representation shift

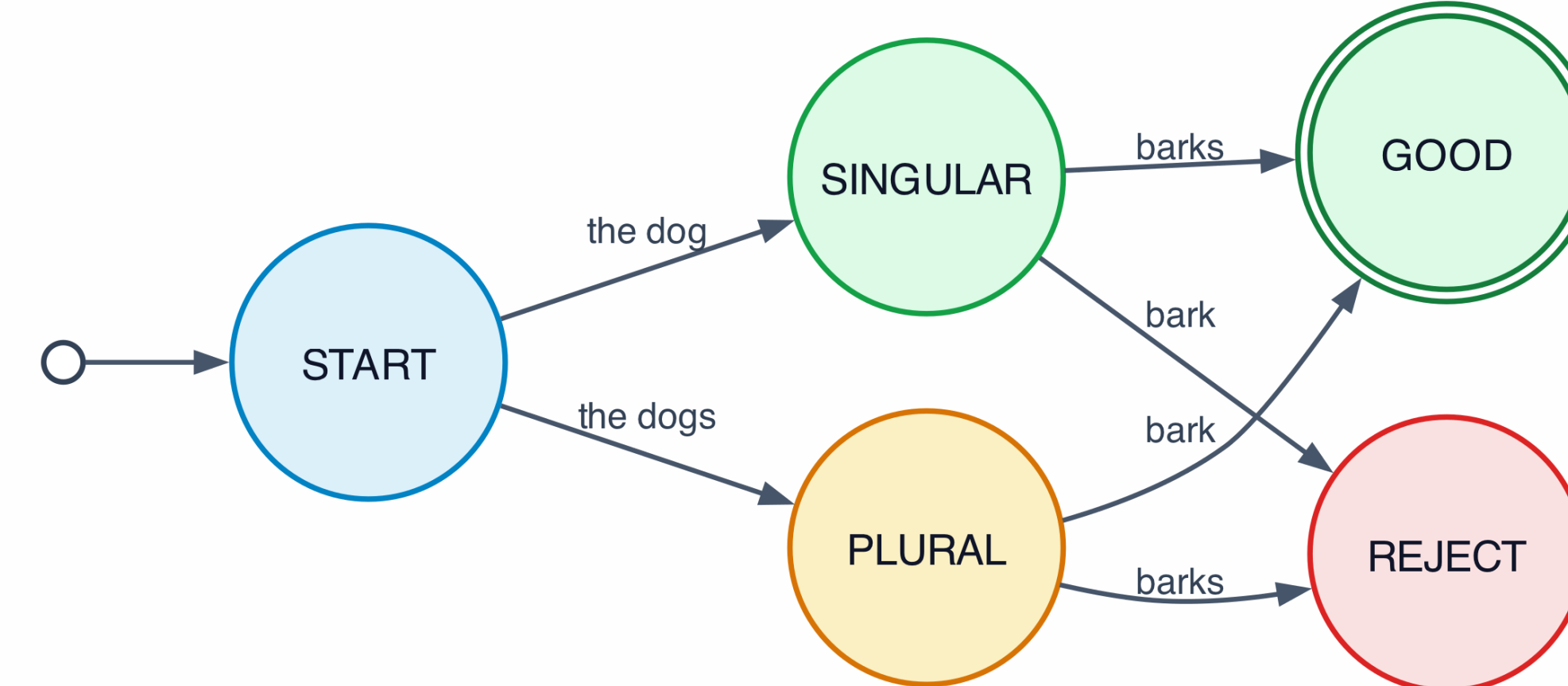
- ▶ Formal grammars describe sentences with clear, rule-based structure, and some languages even have recognizable “shapes” when you look at them with tools from algebraic topology (Ackerman & Cybenko 2021).
- ▶ When we train a neural network on those same languages, it turns them into smooth clouds of points so they become easier to separate, but in doing so it throws away the hidden structure that our proofs depend on.

3 Behavior in real systems

- ▶ Large language models look impressive on benchmarks, but studies show that their behavior can change sharply when inputs contain unusual or noisy language (Miaschi et al. 2022; Oh et al. 2022).
- ▶ They may still assign high probability to clearly bad sentences or struggle on sentences humans find easy, which makes it hard to predict or bound how they will fail ahead of time.

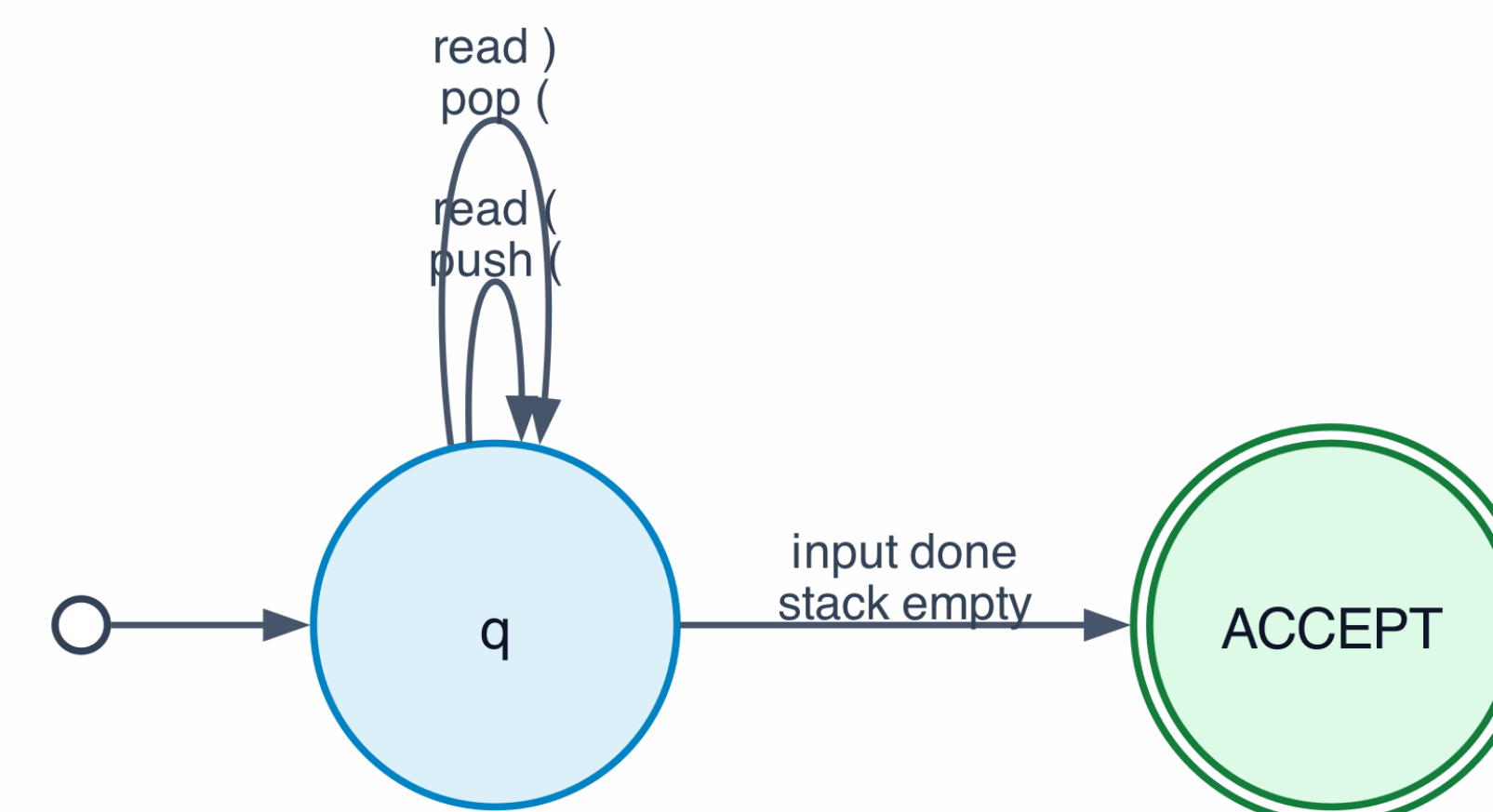
Classical Modeling

Finite-State Automaton for Simple Subject-Verb Agreement



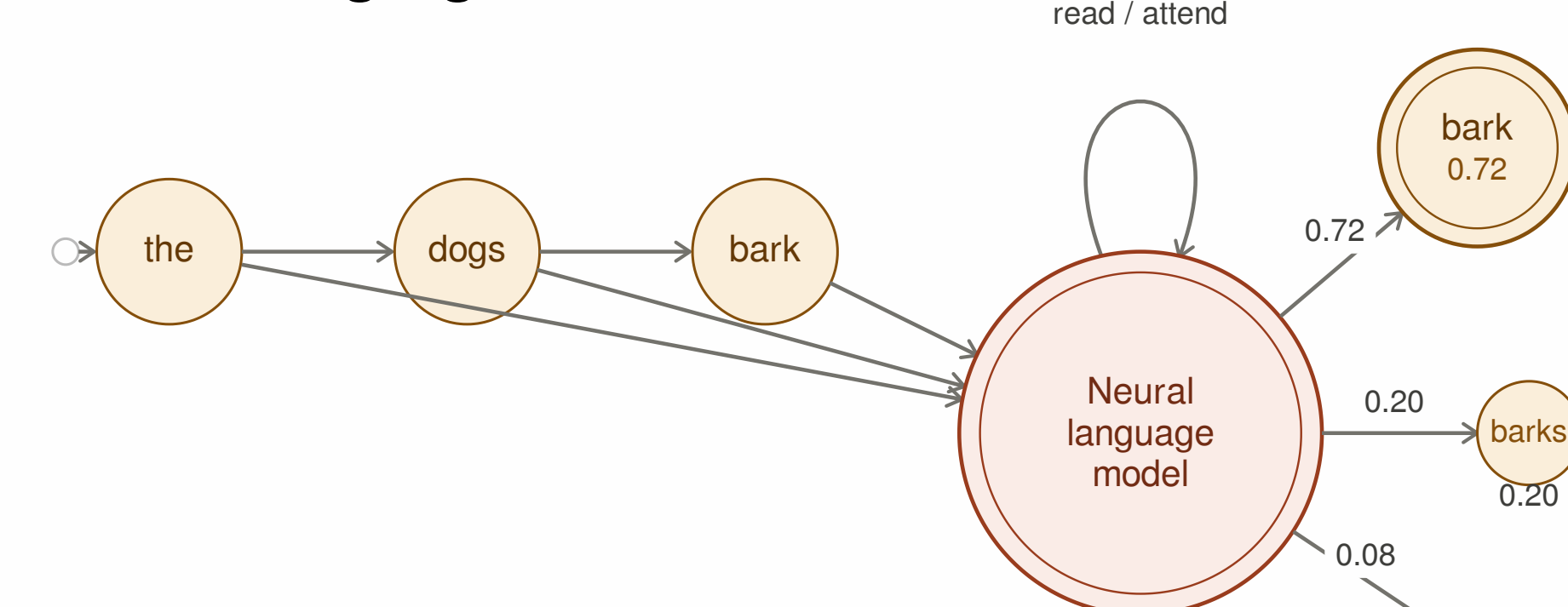
FSA: remembers only the current state – good for simple, local patterns.

Pushdown Automaton for Balanced Parentheses



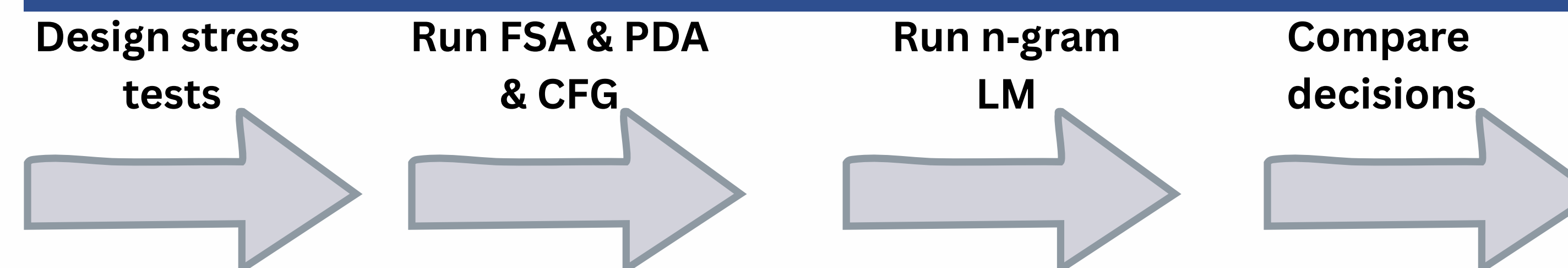
PDA: adds a stack so it can count and match nested items like parentheses.

Neural Language Model for Next-Word Prediction



LM: turns the previous words into a hidden pattern and assigns probabilities to the next word; it never has an explicit “reject this sentence” state.

How we tested this



Patterns:

- ▶ Balanced parentheses at increasing depths (Dyck-1).
- ▶ Two types of brackets, nested and crossed (Dyck-2).
- ▶ Subject-verb agreement, simple and center-embedded (“the dogs that bark see the cat”).

Formal Models:

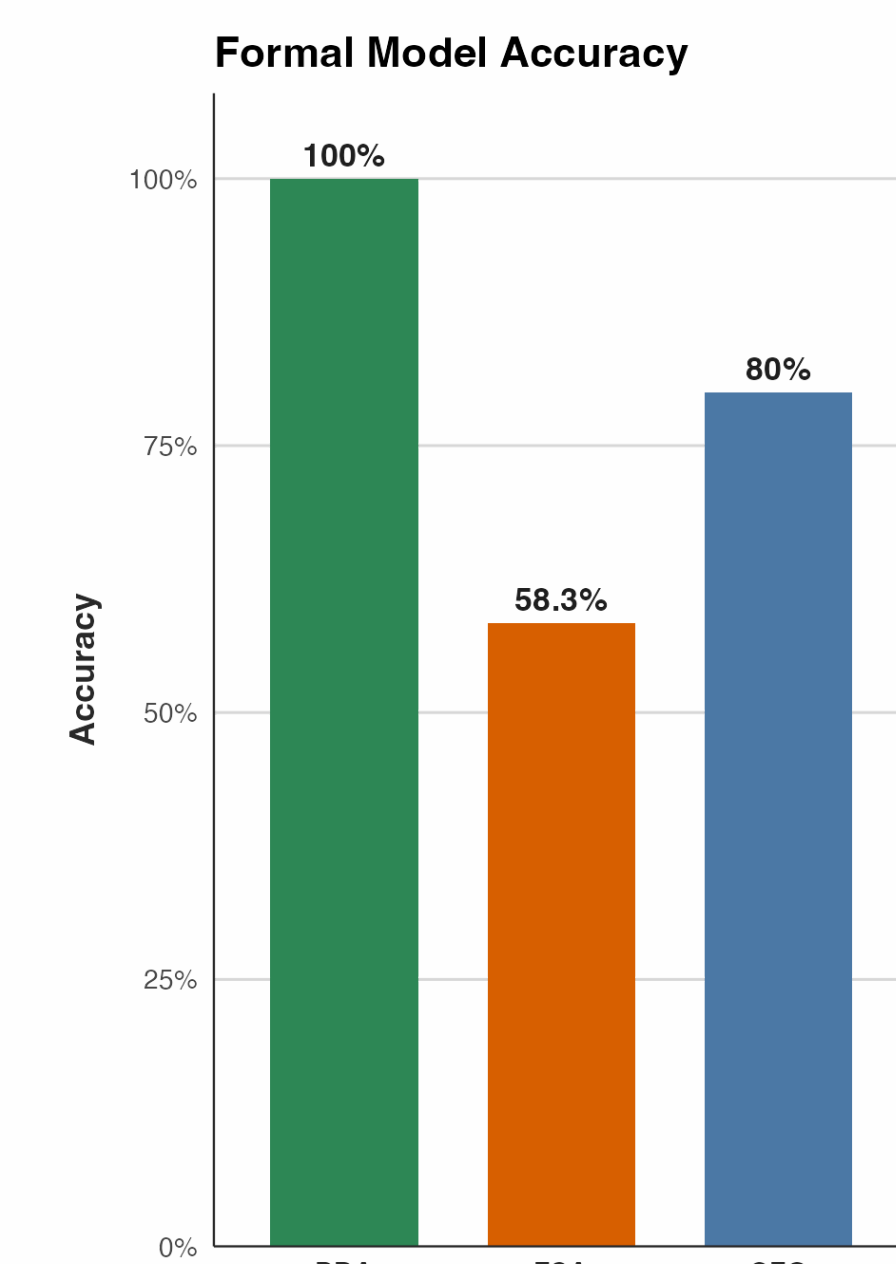
- ▶ Hand-built FSA for shallow parentheses and agreement.
- ▶ PDA for Dyck-1 and Dyck-2; CFG + Earley parser for agreement and embedding.

Language model:

- ▶ Trigram LM with smoothing, as a simplified stand-in for a neural LM; it assigns surprisal but never says ‘reject’.

Results overview

Pattern Type	Pattern Type	LM Result
Dyck-1 (nested parentheses)	PDA 8/8 correct; FSA fails once depth >1	Scores every string; no clear split
Depth scaling	PDA 12/12 across depths; FSA 7/12	Scores change but no hard boundary
Dyck-2 (two bracket types)	PDA 6/6 correct	Valid and invalid strings get similar scores
Agreement + embedding	CFG 8/10; misses only two long-distance cases	LM never rejects; small score gaps



- ▶ Formal parsers draw a **sharp line** between grammatical and ungrammatical strings and maintain it as sentences get more complex.
- ▶ The language model always returns a probability, even for clearly bad sentences, and its scores **often fail to separate good from bad**.
- ▶ When parsers err, we can fix the grammar. When the LM errs, the cause lives in millions of parameters with **no clear repair path**.

Conclusion

- ▶ **Formal grammar parsers still matter:** they offer correctness (no bad strings accepted), completeness (all good strings accepted within the grammar), and verifiability (we can audit the rules). *“Fluent is not the same as provably correct.”*
- ▶ Modern language models give impressive fluency but abandon these guarantees; **their decision boundaries are fuzzy and non-transparent**.
- ▶ Our experiments show this loss even on toy languages, suggesting **the issue is structural, not just ‘we need more data’**.
- ▶ For high-stakes NLP, we argue for **hybrid systems**: a small, provable grammar core surrounded by probabilistic models, rather than grammar-free AI.”

Limitations & Future work

- ▶ **Limitations:** toy grammars, small language model, English-only structures.
- ▶ **Future work:** repeat the stress tests with large pretrained LMs (GPT-2 class), extend grammars to richer center-embedding and tense, and evaluate hybrid designs that enforce parser constraints on LM output.

References

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